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EDUCATION

University of California, San Diego

B.S. Computer Engineering | June 2026

Relevant Coursework: Operating Systems, Networked Services, Databases, Compiler Construction

SKILLS

Languages: Go, C++, C, Rust, Python, Java, TypeScript/JavaScript, SQL, Bash, SystemVerilog, Assembly (ARM, x86)

Backend & Systems: gRPC, PostgreSQL, SQLite, Linux/UNIX, Git, FFMpeg

Graphics: Vulkan, OpenGL, GLFW, GLM

Tools: CMake, Intel Quartus Prime

Other: American Sign Language (biliterate)

EXPERIENCE

Web Engineering Intern | KSDT Radio @UCSD | October 2025-April 2026

- Developed and deployed web application features using TypeScript and React to improve the listener experience.
- Collaborated with cross-functional teams to define requirements, prioritize features, and track development milestones.

PROJECTS

Custom RISC Architecture and Toolchain | SystemVerilog, C, Assembler

- Designed an Instruction Set Architecture for a custom 8-bit RISC processor, defining instruction formats, encodings, and execution semantics.
- Implemented and synthesized a single-cycle CPU in SystemVerilog with Intel Quartus Prime, validating instruction decoding, execution, and memory operations through end-to-end programs.
- Built an assembler that translated assembly programs into machine code while abstracting irregular instruction encodings.

Distributed Video Serving System | Go, gRPC, Protocol Buffers, FFMpeg, SQLite

- Built a concurrent Go HTTP server for video upload, listing, and playback, supporting multiple clients through goroutines and routing content through distributed storage nodes.
- Designed Protocol Buffer interfaces and implemented gRPC communication between the web server and storage nodes for video transfer, retrieval, health checks, and node management.
- Implemented consistent hashing across a dynamic storage ring, tracking predecessor/successor nodes and migrating files when nodes joined or left.
- Developed a Go CLI for launching, terminating, and managing storage nodes, including dynamic file rebalancing; stored video metadata in SQLite.